

Question			Marking details	Marks available				Maths	Prac
				AO1	AO2	AO3	Total		
2	(a)	(i)	Kinetic energy and [gravitational] potential energy referred to (1) $\frac{1}{2}mv^2 - \frac{GMm}{r}$ = total [initial] energy or $\Delta KE = [-]\Delta PE$ (1) Final and initial energy are zero and equal [accept final KE = 0 or final PE = 0] (1)	3			3		
		(ii)	Rearrangement / simplification i.e. $v^2 = \frac{2GM}{r}$ (1) [accept $m = 1$ inserted] Substitution e.g. $v^2 = \frac{2 \times 6.67 \times 10^{-11} \times 1.99 \times 10^{30}}{6.96 \times 10^8}$ (1) Answer = 618 [or 620] km s ⁻¹ (1) c.a.o. [accept ~600 km s ⁻¹ b.o.d.]	1	1 1		3	3	
	(b)	(i)	Applying KE of molecule/atom/particle = $\frac{3}{2}kT$ (or deriving) (1) Rearrangement e.g. accept $v^2 = \frac{3kT}{m}$ (1) Substitution e.g. $v^2 = \frac{3 \times 1.38 \times 10^{-23} \times 5780}{9.11 \times 10^{-31}}$ (1) Answer = 513 km/s (1) [accept just ~500 km s ⁻¹ only with correct substitution]	1	1 1 1		4	3	
		(ii)	Many electrons have enough KE to escape (ecf) or just 'some electrons escape' (1) Because (b)(i) close to (a)(ii) / some electrons have higher velocity / [Boltzmann] distribution / collisions] (ecf) (1) Or protons don't escape		2		2		
		(iii)	Valid method employed e.g. electrostatic force & gravitational forces calculated/equated or other (1) Valid calculation carried out correctly e.g. $F_E = 2.4 \times 10^{-28}$ N and $F_g = 2.5 \times 10^{-28}$ N or charge = 0.084 C (1) [Or using $GMm = kQq \rightarrow 1.2 \times 10^{-10}$ [N m ²] and 1.15×10^{-10} [N m ²]] Valid conclusion (not independent) e.g. she's quite close (1) ecf from calculation of forces [not fields] or alternative above. [Accept: <i>E</i> force bigger / <i>G</i> force smaller]			3	3	2	

		(iv)	Sun electrons $\approx \frac{1.99 \times 10^{30}}{1.66 \times 10^{-27}} \approx 1.2 \times 10^{57}$ (assumption dependent, allow Sun composed of deuterium) (1) [Accept $M_s / 1 \text{ u}$] $\frac{0.08}{e} = 5 \times 10^{17}$ electrons lost or charge on Sun's electrons = $1.92 \times 10^{38} \text{ [C]}$ (1) % lost $\approx 10^{-38}$ (1)		3		3	3	
			Question 2 total	5	10	3	18	11	0